Preprint

Embedded Black Holes in Stars and Planets
e-Anun in the Phraqtal Field

Planetary and Stellar Matter Transformation,
Tension Organization and the Evolution of Planetary Systems

Ukshin Q. Rexhepi

Independent Researcher
Tübingen, Germany

Version 1: October 28, 2026

DOI: 10.5281/zenodo.23735598 [October 2026]

Abstract. This work develops a qualitative interpretation of embedded Anun as central organizational states of planetary and stellar matter configurations within the Universal Quantum Foam Hypothesis (UQSH). An e-Anun is regarded as a field organization that is saturated within its own regime, remains in continuous motion, and can form together with a material envelope and persist over extended periods. The assumed milling function links uptake, the transformation of bound matter, the emission of free tension fronts, and renewed binding in suitable surrounding regions. The possibility that planetary e-Anun can also reorganize raw material accumulated from dust and gas into heavier elemental organizations within their own central surroundings is explicitly examined. The small spatial extent of such a region is distinguished from its local binding and transformative effects. Spatial differences in tension, absorbed radiation, internal motion, and the duration of their action are described as conditions that operate together. In the broader surroundings of a forming star, this perspective is extended to preferred accumulation regions and possible recurring families of planetary configurations. The exposure of an Anun is interpreted as a change in an existing core–envelope configuration. The mathematical connection is provided by the common action, energy exchange, local restoring response, and gravitational source developed in *Light in Light Bubbles*. These existing relations are explained; the new statements concerning element formation and planetary evolution remain qualitative working hypotheses.

Keywords: UQSH; Phraqtal Field; e-Anun; saturation; matter organization; planetary systems; black hole; star; supernovae.

Contents

1 Introduction and Starting Point **4**

2 The Phraqtal Field as a Common Physical Basis **4**

2.1 Naming and the Relationship Among Radiation, Matter, and Gravity 4

2.2 Nested Regimes and Recurring Organizational Roles 5

3 Anun, Embedding, and the Material Envelope **6**

3.1 Name, Saturation State, and External Observability 6

3.2 Iron-Rich Core Regions and the Meaning of the Mantle 7

4 Early Formation, Growth, and Dynamic Maintenance **7**

4.1 Joint Formation of the Core and Envelope 7

4.2 Local Saturation and the Maintained Planetary State 8

5 The Milling Function as a Combined Transformation Process **9**

5.1 Uptake, Breakdown, and the Emission of Free Tension Fronts 9

5.2 Recursive Return Transfer and Conservation in the Overall Balance 9

6 Staged Organization of Matter and Elements **10**

6.1 From Minimal Bound Forms to Atomic Assemblies 10

6.2 Heavier Elements and the Role of the Surrounding Matter 11

7 Planetary and Stellar e-Anun **11**

7.1 Small Spatial Extent and a Pronounced Local Effect 11

7.2 A Planet's Own Transformation of Its Accumulated Initial Material 12

7.3 Stellar Organization and the Maintenance of Complex Environments 13

8 Tension Organization and the Formation of Planetary Systems **14**

8.1 The Planet-Forming Disk Within the Common Field State 14

8.2 Preferred Formation Regions and Local Backreaction 15

8.3 Recurring Configurations Under Comparable Conditions 15

9 Exposure, Envelope Loss, and Cosmic Recycling **16**

9.1 Unveiling an Existing Anun 16

9.2 Distribution of Transformed Matter and Multifunctional Organization 16

10 Stellar Evolutionary Pathways, Continued Embedding, and Exposure **18**

10.1 Progressive Coupling and Different Envelope Responses 18

10.2 The Sun: An Extended Giant Envelope and a White Dwarf Remnant 19

10.3 Core Collapse and a Supernova with Continued Embedding in a Neutron Star 20

10.4 Extensive Exposure, Continued Uptake, and Matter Fallback 21

10.5 Thermonuclear Supernovae as a Distinct Evolutionary Pathway 21

10.6 Shared Conditions and Different Remnant States 22

11 Mathematical Connection to Light in Light Bubbles **22**

11.1 Common Action and Coupled Fields 22

11.2 Mutual Energy Exchange 23

11.3 Bounded Local Restoring Response 24

11.4 Region Growth and the Maintained State 25

11.5 Regime Participation in the Existing Auxiliary Channel 26

11.6 Gravitational Effect of the Overall State 26

12 Spatial Scales and Qualitative Expectations **27**

12.1 Internal Proportions and the Meaning of the Illustrations 27

12.2 Composition, Maintenance, and Recurring System Families 27

13 Synthesis and Further Connections **28**

1. Introduction and Starting Point

The Universal Quantum Foam Hypothesis (UQSH) describes radiation, matter, and gravity as interconnected manifestations of a common underlying field. Within this framework, the question arises of how a highly concentrated central field organization can interact with an extended matter configuration maintained over long periods. To address this question, the present work examines the *embedded Anun*, or e-Anun: an organizational state that is saturated within its own regime and surrounded by planetary or stellar matter. Its envelope is as much a part of the state under investigation as the central region itself. It supplies material and radiation, absorbs emitted field organization, and, through its own motion and binding, influences the conditions under which the e-Anun can persist. This establishes a connection between internal transformation and external persistence that provides the starting point for the following considerations.

The guiding assumption is that such a central organization can arise during the accumulation of dust and gas and evolve together with its surroundings. A planetary e-Anun would therefore, in principle, participate in the same types of binding, reorganization, and stress relief as a larger Anun, while its spatial extent, supply, and total amount of processed matter may differ. This leads to the further hypothesis that a planet can also transform some of its initially accumulated raw material into heavier elemental organizations within its own interior. The e-Anun, its transition region, and the surrounding matter form a coupled setting for this process. In the broader surroundings of a forming star, the same perspective is extended to the spatial distribution of dust and gas: an evolving tension organization could help shape preferred regions of planet formation and favor recurring planetary configurations under similar conditions.

This work develops these ideas as a qualitative UQSH interpretation. Its conceptual basis lies in the works on the Phraqtal Field, gravitational light, and regime-relative saturation; *Light in Light Bubbles* provides the mathematical connection through a common field action, coupled dynamics, mutual energy exchange, and a bounded local restoring response [1–3]. The statements concerning specific element formation and planetary formation zones extend this basis as working hypotheses. Spatial differences in tension, absorbed radiation, internal motion, and the duration of their action are described in relation to one another, without yet deriving specific tension values, elemental yields, or spacing sequences. This qualification applies throughout the qualitative part. It allows a coherent account to be developed while using the existing mathematical results within their actual scope of validity.

2. The Phraqtal Field as a Common Physical Basis

2.1. Naming and the Relationship Among Radiation, Matter, and Gravity

The name *Phraqtal Field* denotes the physically effective organizational form of the common underlying field, nested across scales. The prefix *Ph* refers to its physical and holographically readable manifestation; the lowercase *q* establishes the conceptual connection to the Qu basis, which is not directly measurable. Readability means that an observer perceives the existing organization according to the observer's resolution and coupling to the field. A structure

can therefore appear as a coherent body at one scale, while its internal motion and further forms of binding are resolved at a finer scale. UQSH remains the name of the overarching hypothetical framework. The Phraqtal Field denotes the physical field reality described within it, through whose motion the different physical states take shape [1].

In this interpretation, radiation is described as a free, modulated tension front. Its free local propagation occurs at invariant c; its spatial distribution, superposition with other fronts, and coupling to bound structures can change along its path. The dedensification of a front denotes the altered distribution of its effective organization over a larger region. Matter corresponds to a bound and dynamically maintained field organization whose internal motions remain linked over appropriate time intervals. Its persistence rests on the continuous maintenance of binding and coupling relations. When matter absorbs or emits radiation, or undergoes transformation, the mode of organization of the same physical basis changes within the UQSH framework. This common starting point makes it possible to treat the uptake of material by an e-Anun and the emission of tension fronts as related processes.

Gravity is also part of this overall state. In the existing field-equation work, its source is constructed from the energy and spatial stresses of the fields described and connected to observable motion through a metric formulation. For the present interpretation, this means that the effect of an e-Anun can be fully described only together with its supporting environment. A changed envelope alters the conditions of the core; a changed core acts back on the binding and motion of the envelope. Outgoing fronts and more distant field organization also remain active within the common description. The Anun thus takes on the role of a prominent organizational center within a broader setting whose gravitational effect depends on the entire distribution of energy and stress [3].

2.2. Nested Regimes and Recurring Organizational Roles

UQSH assumes that fundamental modes of organization can recur across nested regimes. A regime denotes a level at which particular structures, motions, and couplings are described together. The finer subregimes support the organization under consideration, while the same configuration can assume the role of a single constituent at a coarser level. This nesting also encompasses different spatial and temporal resolutions. An observer who perceives a body as largely persistent does not thereby resolve every internal motion involved in maintaining it. The recurring mode of organization is therefore described in terms of binding, circulation, exchange, and saturation; the visible similarity of two forms is only a possible indication of such relations.

The analogy between atomlike and galaxylike configurations used in the project works concerns such functional relations. A central region can support binding and centering, while a structured environment permits orbital motion, exchange, and further substructures. The same question is posed for the relationship between an e-Anun and its planetary or stellar mantle: What organization holds the overall state together, what motion distributes absorbed energy, and what pathways permit stress relief or further binding? The triadic perspective treats binding tension, directed motion, and restoring response or exchange as interacting roles. It does not require the body to be divided into exactly three separate objects. Depending on the process, the same spatial region can participate in several of these roles.

An essential clarification concerns the saturation of the supporting regimes. An Anun can reach an upper saturation configuration within its own regime while its subregimes remain close to their ordinary, moving environmental state and carry only an additional change. Participation along the regime sequence is therefore distinguished from the spatial propagation of an effect within a regime. Likewise, the same nested energy must not be counted repeatedly as a new, independent source when the description is refined. This distinction has a direct bearing on the e-Anun interpretation: a highly concentrated central organization is compatible with a continuing carrier organization that has motion of its own. Its embedding can thus be understood as a jointly maintained field state in which the central region and its surroundings perform different organizational functions [3].

3. Anun, Embedding, and the Material Envelope

3.1. Name, Saturation State, and External Observability

The name *Anun* combines several conceptual references. The component *An* consists of the first letter of *Albert* and the last letter of *Einstein*. It also symbolizes beginning and end: what constitutes the upper saturation configuration within one regime can mark the beginning of a new coupling function for a superregime. The *u* refers to universality and UQSH; the final *n* can be read as a regime or dimensional index. The name thus honors Einstein and connects this tribute to the assumed recurrence of physical modes of organization across nested levels. The term derives its physical meaning from the saturation and binding state described [2].

In this work, an Anun denotes a spatially extended, dynamically bound field organization that is maximally saturated within its own regime. The expression *black hole* remains a visual reference for a dark, strongly binding phenomenon; the field-mechanical interpretation investigated here is explicitly developed under the term Anun. The term *embedded Anun* (German: *eingebetteter Anun*), abbreviated *e-Anun*, denotes such a state within a broader matter configuration. The qualifiers planetary and stellar identify its host environment. An e-Anun may already have attained the organizational form characteristic of its regime while remaining surrounded by an envelope whose appearance determines what is observed externally. The visibility of the envelope and the saturation state of the central region are therefore different properties of the same body.

An *Anun precursor*, by contrast, denotes a central organization that has not yet reached the assumed saturation condition. An *exposed Anun* is present when a previously surrounding envelope has largely been shed or altered in such a way that the central state can be observed in a different way. These terms address different questions: how the internal organization is constituted, how it is spatially embedded, and which part of the overall state appears externally. Exposure can therefore involve an Anun that already exists. The same Anun can also assume an *Anektron* role relative to a superregime, provided that its overall organization becomes sufficiently coherent and capable of coupling at that coarser level. Its coupling role in the superregime complements its own saturation; a change in external visibility alone does not determine that role.

3.2. Iron-Rich Core Regions and the Meaning of the Mantle

Two possibilities are distinguished for a rocky planet. An iron-rich central configuration may initially be part of an evolution in which an e-Anun develops under altered binding and supply conditions. It may also constitute the material surroundings of a small e-Anun that already exists. In the second case, the iron-rich material surrounds the saturated field region while remaining part of the larger planetary organization. The term iron core then describes a material region whose internal structure may be finer than its external designation suggests. Whether a particular portion is solid, liquid, or organized in some other way depends on the combined state in question. The hypothesis therefore uses neither iron enrichment nor solidity alone as a marker of Anun saturation.

In the general e-Anun description, the term *mantle* is used relative to the saturated region. It encompasses the surrounding host matter and may therefore include regions that belong to the planetary core in a geological description. This usage is essential to the cutaway view: the small e-Anun, a possible iron-rich embedding region, and the outer layers have distinct spatial extents. The surrounding matter supports compression, internal motion, and exchange. It can transfer absorbed radiation, carry material into the central region of influence, and transport portions transformed there back to greater distances. Its backreaction on the e-Anun thus arises from its entire organization, including its capacity to absorb energy, maintain bonds, and redistribute altered stresses spatially.

Within this account, the iron-rich embedding region can contain both previously accumulated matter and matter that has undergone further transformation within the body itself. The mantle thus acquires a dual role: it is part of the conditions under which the e-Anun persists and a possible carrier of the products of its transformation processes. This relationship connects the origin of the initial material to the subsequent evolution of the body. An image of the planetary interior should therefore show a jointly supported structure in which the central region, its immediate material surroundings, and the outer layers remain interconnected. The details of their composition are also shaped by transport, the duration of the processes acting on them, and their evolutionary history. A small saturated region can thus be embedded within a much larger material configuration maintained over long periods.

4. Early Formation, Growth, and Dynamic Maintenance

4.1. Joint Formation of the Core and Envelope

The formation of an e-Anun is considered a possible part of the early accumulation and reorganization of matter. As dust, gas, and absorbed radiation contribute to a common configuration, more concentrated states of binding and tension can develop in the interior. If supply, spatial closure, and mutual coupling take a suitable form, a central region can approach saturation within its own regime. Its subsequent evolution occurs in an environment that is simultaneously growing, moving, and changing its own distribution. The eventual planet or star is thus described as the result of a joint evolution in which the central field state and the envelope influence one another from the outset. In this picture, Anun formation need not be delayed until a late event involving envelope loss.

An initially very small saturated region can already play a pronounced local role, even though it occupies little space relative to all the accumulated matter. Whether its region of influence grows depends on how the surrounding configuration permits further uptake and restoring response. Material supply alone therefore does not fully determine the outcome: the direction of motion, the envelope's absorption of outgoing fronts, and the available pathways for release also contribute to the evolutionary course. A persistently suitable supply can expand the bound region spatially. Under other conditions, additional uptake may predominantly lead to exchange or stress relief. Different evolutionary paths can thus be described within the same basic field logic without assigning a particular future host form to every early core.

4.2. Local Saturation and the Maintained Planetary State

Saturation and spatial growth refer to different aspects of this evolution. The local restoring response can approach a limiting range while the extent of the organization involved changes. Additional compatible uptake can then expand the saturated region, its transition structure, or the distribution of its energy fluxes. The notion of maximal saturation within a regime therefore does not necessarily imply that the entire body can no longer absorb energy. For the evolution of the envelope, the decisive issue is how this uptake enters the combined state. A larger e-Anun may be accompanied by an altered envelope organization; alternatively, a new balance may arise in which the spatial extent is maintained on average while exchange continues.

Such a maintained state is termed a *maintained planetary core*. Here, maintained means an organization that persists over a suitable period while internal motion continues. Uptake, return transfer, and release can balance one another so that neither the extent of the e-Anun nor its bound energy increases substantially on average. Its envelope provides the environment in which this balance is possible. A nearly constant external appearance is therefore compatible with internal processes in which matter is repeatedly reorganized and radiation is absorbed or emitted. The body's constituents jointly support a particular form while the motion involved in maintaining it continues. Whether this form is regained after a perturbation is the further question of its dynamic stability.

Within the hypothesis, the maintenance of a planetary state allows for long periods of evolution. The mere passage of time need not cause a body to become stellar or to develop an ever-increasing external gravitational effect. Redistribution of material between the envelope and the central region likewise does not, by itself, increase the body's total energy. It can, however, change the body's internal tension distribution, its exchange with the surroundings, and the pathways of further transformation. External persistence and internal evolution are therefore considered together. The notion of a maintained e-Anun specifically allows for a continuing internal process whose extent is limited by supply and backreaction and whose material products can accumulate in the envelope over extended periods.

5. The Milling Function as a Combined Transformation Process

5.1. Uptake, Breakdown, and the Emission of Free Tension Fronts

The term *milling function* illustrates the assumed ability of an Anun environment to convert absorbed matter into other organizational forms of the same field. Incoming material carries existing bonds, internal motion, and energy. If it enters a region whose tension and exchange conditions differ substantially from those of its previous environment, these binding relations can be altered or partially disrupted. One portion can remain bound and become part of the central or surrounding organization; another can be converted into freely propagating tension fronts. The transformation thus concerns the relations through which a material form is maintained. The mill metaphor illustrates this connection and draws attention to the repeated passage of existing organization through altered binding conditions.

The process encompasses the Anun, the transition region, and the associated envelope. The organization of incoming material can change as it approaches, while outgoing fronts encounter bound matter again in the same surroundings. This produces a spatially structured exchange in which uptake and release can occur simultaneously at different locations. Maximal saturation within the regime then signifies a particularly strongly bound state that can respond to additional stress through several pathways: spatial expansion, altered internal organization, participation of supporting regimes, or front emission. Which of these possibilities predominates depends on the combined configuration. An envelope that reabsorbs a large portion of the emitted fronts opens different exchange pathways from an environment through which those same fronts largely continue outward.

In the UQSH account, high-frequency tension fronts are considered possible carriers of intensive reorganization. Their significance arises from their actual energy organization, spatial superposition, and coupling to the surroundings. For a superposition to favor a bound response, its direction, phase, and duration must be suitable; the mere presence of several fronts does not establish amplification. The existing field-equation work provides relations among encounter geometry, internal orientation, spatial extent, and the duration of an effective excitation [3]. For the milling function, this yields a picture that connects to both the conceptual and mathematical framework: breakdown and renewed concentration can be parts of the same exchange process, provided that the environment in question supports the corresponding forms of reorganization.

5.2. Recursive Return Transfer and Conservation in the Overall Balance

The envelope can reabsorb outgoing fronts and incorporate their energy into its own motion or binding. Further fronts can then propagate back, excite other regions, or reach the central region of influence again together with transported matter. An initially absorbed matter configuration can therefore participate repeatedly in different transformation and binding processes. Recursive, in this context, means that the results of one process become conditions for subsequent processes. Exchange changes the surrounding matter, which then provides an altered environment for the next uptake. In this way, the entire body can undergo internal

evolution even while individual regions retain similar external properties over extended periods.

Here, *fresh matter* means newly bound or newly assembled field organization. Its formation uses existing energy, already bound forms, and exchange with the surroundings. The milling function therefore describes redistribution and reorganization within a common balance. If energy is released outward over an extended period, that release must be sustained by a corresponding supply or by a change in the stored organization. The mathematical connection in Section 11 makes this reciprocity explicit: the exchange that increases one specified field contribution appears with the opposite sign in the other contribution. This relation supports the notion of continuous internal transformation while also limiting its possible duration and strength through the actual supply to the overall state.

6. Staged Organization of Matter and Elements

6.1. From Minimal Bound Forms to Atomic Assemblies

The first stage of the proposed organizational pathway comprises minimal bound field forms capable of coupling. Previous UQSH works consider electronlike modes and modes with complementary orientations as possible manifestations of such binding [1, 2]. The crucial point for the present account is that a free front and a persistent material form differ in their entire internal organization. An excitation must remain held together beyond its immediate passage, establish suitable relations among its components of motion, and respond to perturbations with a restoring response that maintains it. The formation of such a form therefore requires more than a high instantaneous tension value. Spatial closure, mutual coupling, and a supporting environment must operate together if a transient excitation is to become a longer-lasting organization.

At the next qualitative stage, hydrogen and helium are considered the first simple families of atomic organization. This transition involves the development of suitable nucleuslike binding and its joint organization with the other atomic components. In the UQSH interpretation, electronlike and nucleuslike roles, as well as roles that act neutrally, are understood through their relations within a bound configuration. The shorthand statement that simple elements arise from initial electron forms is accordingly elaborated as a reference to a multistage organizational pathway. Further binding changes the structure of the entire assembly. Which form persists depends on whether its internal motions, coupling orientation, and exchange with the surroundings are mutually compatible. The assumed sequence thus describes organizational stages whose specific material identification concerns the entire configuration in each case.

This process connects the smaller bound forms to the broader environment of an e-Anun. Fronts that arise in a region of intense transformation can encounter existing bound structures in other regions. Their absorption there can enable new relations, excite existing forms, or disrupt them again. The environment thus performs a selection function in the physical sense: under particular conditions, some organizational forms persist, while others appear only briefly or are reorganized again. Which forms accumulate over extended periods therefore depends on both their formation and their maintenance. This distinction leads to the question

of the spatial and temporal conditions under which heavier elemental organizations can be produced.

6.2. Heavier Elements and the Role of the Surrounding Matter

The formation of heavier nuclear and atomic organization is associated with an environment in which additional binding relations can be established and maintained for a sufficient time. Gravitational centering retains material within the common region of influence, while local differences in tension, restoring response, and energy exchange affect its further organization. A region of intense breakdown can therefore lie adjacent to a region in which more complex bound forms persist. Proximity to the e-Anun describes only part of these conditions. The existing matter, its direction of motion, the absorption of radiation, and the duration over which a particular configuration is effective are equally significant. This work identifies these factors together so that the assumed element formation can be understood as a process structured in both space and time.

The term *categorical organization* means that different binding and maintenance conditions can favor different families of persistent forms of matter. A category is thus determined by the type of organization that is possible and by its environment. This idea does not require a rigid ordering of all elements along a single radius. Internal motion can carry material from one region to another, where it undergoes further transformation and subsequently persists in a quieter environment. The material composition of a mantle region therefore records the history of several processes: the accumulation of the initial material, its subsequent excitation, the possible formation of new bound forms, and their later transport. The spatial arrangement of the conditions and the subsequent spatial distribution of their products must therefore be considered together.

In this interpretation, the mantle serves functions of formation, maintenance, and storage. It can supply material to the central transformation region, absorb outgoing fronts, and support newly bound forms over extended periods. Its backreaction, in turn, changes the supply and external conditions of the e-Anun. The material envelope is thus an active part of the assumed elemental organization. In particular, the possibility of preserving the products of intensive internal reorganization outside the region that directly disrupts them connects the milling function to the formation of heavier matter. How much of this matter remains in the body or is later released depends on the subsequent evolution of the envelope and its exchange with the surroundings.

7. Planetary and Stellar e-Anun

7.1. Small Spatial Extent and a Pronounced Local Effect

A planetary e-Anun is understood as the same basic type of saturated field organization as a larger Anun. Here, the term *deep gravity* denotes the highly concentrated organization of gravitational binding and tension in its central region. A small spatial extent can coexist with a pronounced local effect on matter and energy. This idea requires a distinction between the strength of a local effect and the amount of material affected by it. A small e-Anun

can have an intense transformation region while only a limited portion of the surrounding matter participates in it at any one time. Regions farther out can experience conditions that maintain their bonds over long periods. The scale of the entire body is thus determined by the interaction of these different regions.

The common type of organization allows for different specific tension profiles, exchange times, and rates of energy transfer. A planetary and a stellar e-Anun can therefore differ considerably in their spatial extent, supply, and the amount of matter transformed in each case. Their comparability rests on the assumed relationship among saturation, binding, and reorganization. It does not require all local tension values to be equal. This is particularly important when depicting a small core within a large planet: a small drawn size is intended to show the limited spatial extent of the saturated region, while its combined effect with the surroundings is conveyed through the surrounding layers and their relationships. Figure 1 illustrates this distinction between internal extent and the encompassing host configuration.

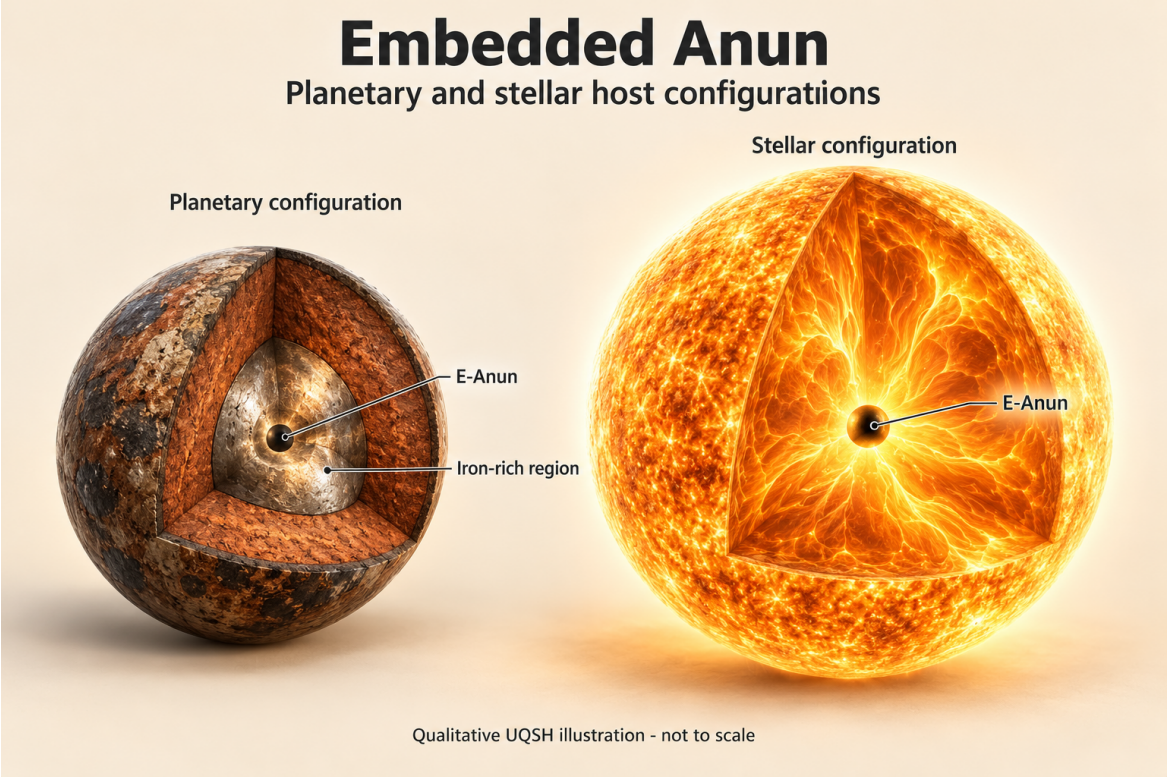


Figure 1: Planetary and stellar host configurations with an embedded Anun. The cutaway exposes a small central saturated region and its surrounding material; the planetary example includes an iron-rich region. Host sizes, core sizes, and internal proportions are schematic. Conceptual UQSH illustration; no physical radius ratio has been calculated.

7.2. A Planet's Own Transformation of Its Accumulated Initial Material

Under the assumption developed here, a planetary e-Anun can also transform raw material initially accumulated from dust and gas within its central surroundings, thereby producing heavier elemental organizations. The planetary body is thus assigned an internal evolution of

its own, in which the origin of the initial material is connected to its subsequent reorganization. The accumulated forms already possess binding and motion. If they enter the inner region shaped in part by the e-Anun, their couplings can change and portions can be transferred onward as free tension fronts. In suitable adjacent regions, surviving bound forms and reabsorbed fronts can subsequently participate in further binding. The planetary e-Anun thus takes an active role in the evolution of its host's material composition.

The proposed process encompasses the combined configuration of the saturated region, transition region, and surrounding matter. Gravitational centering retains material within this coupled setting, while local conditions determine which forms can persist there. A portion of material may initially enter a region of stronger transformation and, after its binding changes, return to a different environment. There, it may persist over an extended period or participate in exchange again. Previously accumulated elemental forms and those newly reorganized in the interior can therefore jointly shape the evolution of the core and mantle. The hypothesis thus allows for a contribution to elemental organization from within the planet, whose significance depends on the actual supply, the extent of the regions involved, and their evolutionary history.

Spatially varying tension and exchange conditions are expected around the e-Anun. Intensive reorganization and front emission may predominate in its immediate vicinity; adjacent regions may allow further-developed bonds to persist for longer. Distance, direction, and local matter organization jointly determine this spatial structure. Absorbed radiation changes the exchange available, internal motion connects different regions, and the duration of their action helps determine which changes can accumulate. Field tension, energy density, and matter density are treated as coupled quantities that must each be described separately. The expected spatial tension organization concerns their combined distribution and therefore cannot be reduced to a single density value.

An e-Anun maintained on average can remain involved in this process without continuously increasing its spatial extent. Its persistence allows repeated uptake and return transfer to involve the same central organization over extended periods. The material surroundings can receive the products and, in turn, alter subsequent exchange. In this interpretation, an iron-rich embedding region would be both part of the conditions and part of the body's evolutionary history. The decisive requirement is the common balance: strong local transformation and a largely persistent external planetary appearance must be compatible through supply, transport, and release. The account thus connects an active central process with the possibility that outer layers support persistent material and chemical structures over long periods.

7.3. Stellar Organization and the Maintenance of Complex Environments

A stellar configuration is considered under conditions in which supply, internal motion, and energy exchange support an extended radiating envelope. Within it, the assumed e-Anun forms an organizational center whose effect is described together with the stellar matter. The star's appearance arises from the distribution of binding, transformation, transport, and front emission throughout the body. Processes observed and described as fusion and elemental transformation are interpreted within UQSH as particular forms of reorganization of bound

field states. The qualitative account connects these processes to the question of which spatial conditions support them and how their energy transfer contributes to the continuing stellar organization.

The differences from a planetary state therefore involve more than the body's visible size. A different supply and a different envelope organization can cause larger quantities of material and stronger energy fluxes to participate in exchange. Internal patterns of motion can also change, causing certain regions to encounter emitted radiation and transported matter more frequently or for longer periods. A possible increase in disordered or varying motion is treated as part of this joint evolution. A larger saturated region, a larger radiating envelope, and a higher rate of energy conversion remain distinct properties whose relationship depends on the particular state. The terms planetary and stellar thus denote different overall configurations within the assumed common mode of organization.

For complex chemistry and environments capable of supporting life to be possible, intensive internal transformation must be able to coexist with sufficient persistence in other regions. Within the hypothesis, maintained material bonds and a sustainable energy exchange provide conditions under which more complex structures can develop further. An e-Anun alone does not determine whether these structures arise; its significance lies in its assumed participation in the combined binding and transformation processes. The spatial structure of a body and its surroundings thus becomes an essential part of the account. It connects the strong local effect of the central region to the possibility that different timescales, different exchange conditions, and persistent material forms become relevant farther out.

8. Tension Organization and the Formation of Planetary Systems

8.1. The Planet-Forming Disk Within the Common Field State

The formation of a star from accumulated dust and gas is also considered an evolution of its surrounding field organization. As the central matter configuration becomes denser and its assumed e-Anun develops, the binding and exchange conditions in its surroundings also change. The existing material has a spatial distribution and motion, absorbs radiation, and contributes to the overall state through its own energy and tension. From the outset, the forming star is therefore connected to an environment that is already active. This environment provides conditions for the star's evolution and is itself changed by the growing central organization. In this account, the eventual distribution of planets is understood as part of that joint evolution.

The field organization remains spatially three-dimensional. Rotation and directed material supply can establish a preferred plane in which material accumulates particularly strongly and remains mutually coupled over extended periods. An accretion-like, planet-forming disk is thus interpreted as the material expression of a jointly developed organization of field and motion. Its visible form reveals part of this organization, while further field effects can extend beyond the dense material plane. The disk-like appearance is therefore associated with the existing motion and the spatial distribution of supply. The star, its assumed e-Anun, and the disk matter act back on one another, continuously changing the tension distribution as the system forms.

8.2. Preferred Formation Regions and Local Backreaction

Regions with different tension strengths, spatial variations in tension, and matter densities can develop within this environment. In some regions, material may primarily continue to be transported, while in others it remains bound for longer or accumulates repeatedly. Absorbed radiation, internal motion, and the duration of the respective conditions help determine whether a transient accumulation can become a more persistent configuration. The approach therefore connects preferred formation regions to the combined behavior of these quantities. A single high tension or density value is no more sufficient for this description than a distance from the star alone. Over a sufficient period, the local environment must support an appropriate form of binding, supply, and restoring response.

For distinct formation zones or ringlike accumulations, an additional assumption is introduced: the coupled motion may persistently or repeatedly favor particular regions. The general spatial decay of a central effect does not, by itself, determine the number or location of such regions. The assumed structure emerges only through the interaction of supply, motion, exchange, and local binding. It may shift, strengthen, or weaken again during evolution. This qualitative account therefore refers to preferred regions whose actual occupation depends on subsequent evolution. This allows similar initial conditions to produce recognizable common features while individual accumulations grow differently or alter their surroundings.

Forming planets act back on this organization. Their accumulated matter changes the local gravitational and dynamic configuration, and any e-Anun of their own may support further reorganization in their interiors. Evolution within the planetary body and evolution of the broader system thus interact. A change in material supply affects the young planet; its increasing binding and motion, in turn, affect the subsequent distribution of material. The planet's initial material is therefore shaped both by its formation environment and by its later internal history. This reciprocity is essential to the UQSH interpretation because it places the individual planetary organization within the common field state of the stellar system.

8.3. Recurring Configurations Under Comparable Conditions

In this approach, recurring families of planetary configurations are expected under similar formation conditions. These conditions are characterized by the mass, extent, and evolutionary stage of the central star, the organization of its assumed e-Anun, the available material, and that material's motion. They also include radiation absorption, the evolution of supply over time, and influences from the broader surroundings. The visible size of the star may be a feature of the combined configuration, but it describes that configuration only partially. Two systems with comparable external stellar sizes can have different histories, while systems of different absolute sizes can be similarly organized in certain relations among their binding and exchange conditions.

The assumed recurrence therefore concerns, in particular, the relations within a system: which regions preferentially accumulate material, which scales emerge together, and how distances relate to one another. Similar relationships among the relevant conditions could favor similar planetary size ratios or spacing sequences. This does not establish a fixed proportional enlargement of all orbits and bodies. Selecting an appropriate comparison scale is part of the subsequent description of this expectation. Differences in supply, motion, and

the duration of their action can produce a variety of individual configurations within such a family. The hypothesis thus connects the recurrence of basic organizational possibilities to the individual evolution of each planetary system.

9. Exposure, Envelope Loss, and Cosmic Recycling

9.1. Unveiling an Existing Anun

Unveiling denotes the loss or extensive restructuring of a material envelope that changes how a previously embedded Anun can be observed. In the picture developed here, the central saturated state may have existed before this external change. The evolution leading to exposure then primarily concerns the combined binding and supply conditions of the core and envelope. When these conditions change, portions of the surrounding matter may lose their previous organization, be released outward, or take on a different arrangement. The point at which the central region becomes more apparent thus marks a change in the overall configuration and its observability. Figure 2 schematically depicts this possible transition.

An Anun precursor, an embedded Anun, and an exposed Anun therefore do not constitute a mandatory age sequence for every body. An e-Anun can remain within a planetary or stellar envelope maintained over long periods. The envelope can also undergo partial restructuring while the central region remains embedded. Even during exposure, a distinction must be made between changes in the Anun's own organization and changes only in its relationship to the remaining surroundings. The schematic illustration therefore retains a recognizable central region throughout the sequence. It is intended to convey that observable unveiling can reveal an already existing internal state, while the actual size and energy of that state during the transition would have to be determined from the coupled dynamics.

9.2. Distribution of Transformed Matter and Multifunctional Organization

In this account, strong outbursts and supernova-like envelope rearrangements are considered possible processes through which bound matter and free tension fronts enter the broader surroundings. Matter that was previously accumulated, transformed internally, or maintained in the mantle can thus participate in new accumulations. The composition of the released material then carries the body's evolutionary history. Its release connects the internal transformation process to the external distribution from which later configurations can arise. The duration of the preceding processes and transport between different internal regions thereby also become relevant to the material available in the surroundings after a change in the envelope.

The assumed multifunctionality of the Anun lies in the connection among these processes. It participates in centering and binding, contributes to the reorganization of absorbed matter and to front emission, and changes the conditions of its surroundings. In this view, embedded and exposed forms belong to the same basic field logic, while their specific exchange pathways differ. The envelope can return some of the emitted organization or store it over extended periods; after the envelope changes, other outward pathways can become active. The cosmic

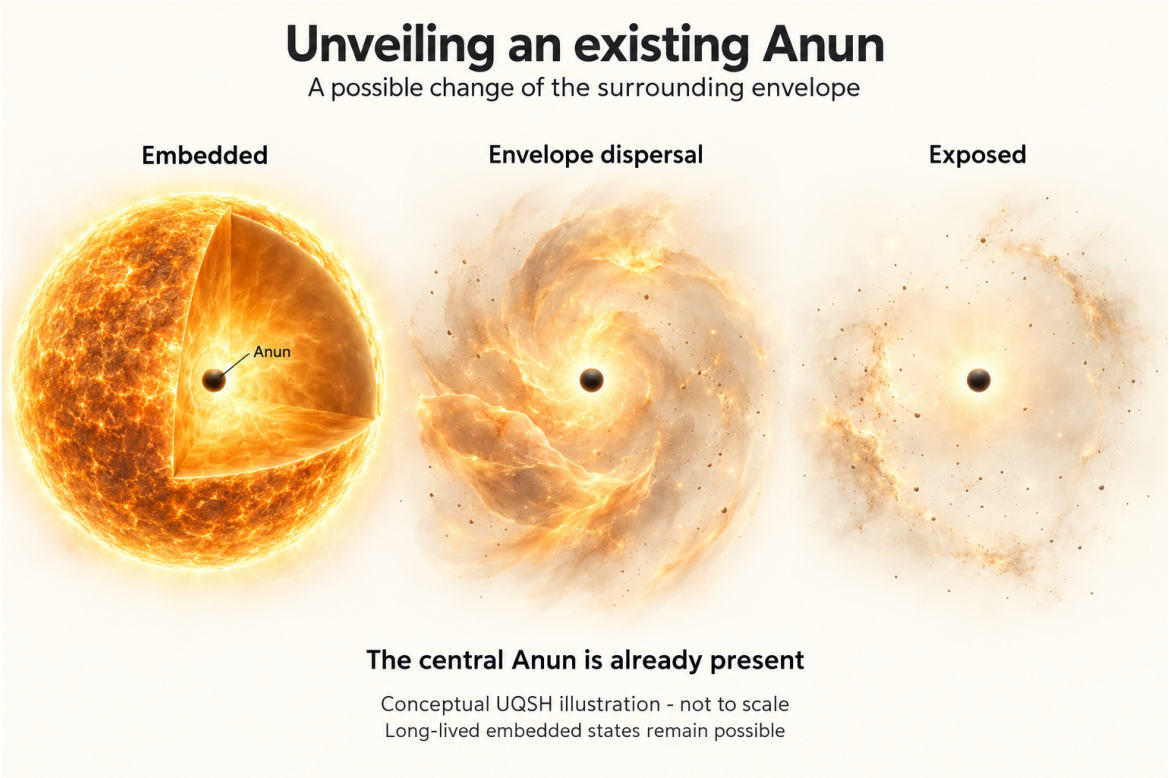


Figure 2: A hypothetical transition from an embedded to an exposed Anun through envelope dispersal. The repeated central symbol indicates that the Anun may exist before the envelope changes; it does not impose a constant physical core size during the transition. Long-lived embedded states remain possible. Conceptual UQSH illustration; not to scale.

setting thus appears as a continuous reuse of field organization in which uptake, binding, transformation, and distribution repeatedly provide conditions for one another.

This picture connects the local evolution of a body to broader contexts without assigning the same temporal course to every individual configuration. A planet maintained over long periods, an active stellar envelope, and an exposed Anun can represent different forms of the same assumed interplay between binding and exchange. Fresh matter again denotes newly bound or newly assembled organization formed from existing field energy. Its formation and release must be described within the common balance of the system in question. On this basis, the qualitative idea of a continuously reorganizing universe can be formulated, in which the persistence of individual forms and changes in their surroundings operate simultaneously.

10. Stellar Evolutionary Pathways, Continued Embedding, and Exposure

Within the UQSH interpretation examined here, stellar evolution is viewed as the joint evolution of a star's assumed e-Anun, the surrounding matter, and their exchange relationships. The central saturation state alone determines neither the star's eventual appearance nor the manner in which its envelope might be lost. A spatially extended Anun that is saturated within its own regime can be associated with different material environments. Its local organizational form, the extent of its effective coupling region, and the maintenance of its envelope must therefore be described separately. This distinction follows the preceding treatment of saturation, redistribution, and common field effects [2, 3].

The evolutionary pathways distinguished in astronomy provide reference points for the following qualitative description. Not every star undergoes a supernova, and not every supernova leaves the same remnant. Assigning an e-Anun to the interior of a star, white dwarf, or neutron star is part of the hypothesis developed here. The cited studies of stellar evolution describe the respective evolutionary pathways; they do not establish the assumed embedding of an Anun. The initial aim is to connect these different phenomena conceptually within UQSH without attributing a uniform sequence or an already calculated internal mechanism to them.

10.1. Progressive Coupling and Different Envelope Responses

The starting point is the assumption that an already saturated e-Anun can alter its exchange relationships with the surrounding matter under continued supply. Coupling inhibition concerns further uptake into the same central form of binding. Energy and matter that cannot remain incorporated in this form under the prevailing conditions can be transformed, emitted as radiation, or transferred through internal motion. If these contributions are absorbed by additional regions of the envelope, those regions may participate more extensively in the common exchange. Such an expansion of the effective coupling region does not require an abrupt change in central saturation and must not be equated with an expansion of the saturated core by the same amount.

The envelope can respond to this evolution in different ways. It can alter its internal motion, expand spatially, shed individual layers, or form a new compact state after a more substantial rearrangement. The possible response depends on its binding, composition, supply, and available transport pathways. Spatial differences in tension, absorbed radiation, internal motion, and the duration of their action jointly determine how a change spreads through the body. The released energy need not remain entirely within the envelope. One portion may reach the surroundings, while another is absorbed within the envelope and influences its subsequent evolution. Only this actual interaction allows a passing front to contribute to a change in envelope organization.

In this interpretation, a possible tipping point concerns the maintenance of the combined core–envelope configuration. As long as uptake, redistribution, and release remain compatible, the change can proceed gradually or recur. If the existing organization of binding and motion can no longer be maintained, however, an initial outward motion could promote further redistribution and release. A development prepared over a long period can then trigger a comparatively rapid visible response. An increase in the coupling region alone is not sufficient: the decisive issue is whether its backreaction continues to support the existing envelope, brings it into a different bound state, or enables it to detach. Envelope ejection does not require a general reversal of gravity.

10.2. The Sun: An Extended Giant Envelope and a White Dwarf Remnant

Current stellar evolution models predict that the Sun will evolve through giant phases and lose large parts of its outer envelope before becoming a white dwarf. This pathway does not involve a supernova. More generally, stars of low and intermediate initial mass can release material into their surroundings through winds and mass loss associated with pulsations during their late evolution. A compact material remnant remains and subsequently cools over a long period [4, 5]. For the UQSH interpretation, the essential point is that a substantial expansion of the visible envelope must be compatible with the persistence of a much smaller inner region.

From a field-mechanical perspective, the giant phase can be described as a possible transition to a more extended envelope organization. A slow change in internal transformation and supply could shift the regions in which radiation is absorbed, motion is excited, and binding is maintained. The envelope would then respond through redistribution and expansion, while inner material regions could simultaneously become denser. Under the assumption of an e-Anun, its altered coupling to the envelope would be part of this joint process. The increasing stellar radius would indicate a change in the host configuration; it would not directly measure the radius of the saturated Anun. Any expansion of the effective coupling region would also have to be assessed separately, because a geometrically larger envelope need not participate equally strongly in exchange at every location.

Such an evolution can pass through new maintained states over an extended period. Repeated changes in energy absorption and internal motion could affect individual outer regions in ways that cause them to expand or detach while other regions remain bound. Envelope loss would then be distributed over several events. Within the hypothesis, this corresponds to continued relief and rearrangement whose spatial extent is limited by the binding conditions

at each stage. A single explosive tipping point is not required. At the same time, each layer that is shed changes the conditions of the remaining body, so subsequent exchange does not simply reproduce that of the original star.

In this interpretation, the white dwarf would represent a compact remaining organization of matter. If an e-Anun was present in the progenitor star and persists, it could remain embedded in this matter. The loss of the extended giant envelope would then not constitute complete exposure of the Anun. Relative to the Anun, even the compact remnant matter would still provide an embedding environment. Long-term cooling requires a corresponding evolution of the overall balance: a local saturation form does not imply an indefinitely constant rate of energy transformation. With a change in supply, the rate of processing can decrease while the bound organization persists. Within the hypothesis, the Sun could thus undergo a transition from an extended radiating host configuration to a more compact embedding environment that progressively cools.

10.3. Core Collapse and a Supernova with Continued Embedding in a Neutron Star

In another evolutionary pathway, the inner region of a massive star loses its previous support and undergoes dynamic collapse. An outward explosion can eject large parts of the overlying matter while leaving a neutron star as a compact remnant [6]. The inner collapse and the outward release involve different regions and directions of motion. The UQSH interpretation must preserve this distinction. A possible tipping point in the envelope must therefore not simply replace all internal processes; instead, the description must address how the altered internal organization and the energy transferred outward could influence one another.

Under the assumption of an existing e-Anun, the collapse could initially involve the matter surrounding it and that matter's coupling. Its previous spatial organization would no longer be maintained, causing motion, tension, and exchange to change rapidly. The effective coupling region of the central state could thereby be restructured without the central state itself first having to undergo an abrupt jump in saturation. Successful envelope ejection would require a sufficient share of the available energy and directed motion to reach the outer regions and enable their detachment. The matter remaining farther inward, by contrast, could develop a new, substantially more compact form of binding and maintenance.

In this picture, the neutron star would be a possible dense host configuration for an Anun that remains embedded. As elsewhere in the manuscript, the term *mantle* denotes the matter relative to the assumed Anun and is not equivalent to a particular geological or astronomical layer designation. The remaining embedding environment would also not simply be an unchanged portion of the former stellar envelope: it would have undergone substantial reorganization of its own. Its maintenance would have to arise from the combined state of binding, spatial tension, motion, and energy exchange. An outburst in the outer regions and continued central embedding would therefore be compatible. The supernova would not necessarily expose the Anun completely, even though a large part of the original host matter had been released.

10.4. Extensive Exposure, Continued Uptake, and Matter Fallback

Another possible course concerns a configuration in which no material embedding environment of the white dwarf or neutron star type persists over the longer term. In the UQSH picture examined here, the Anun could then appear largely exposed relative to the former stellar envelope. This means that the previous host matter no longer determines its external appearance in the same way. The term denotes neither a visible solid surface nor the complete absence of surrounding matter. An exposed Anun can continue to interact with gas, a disk, or outflowing material. Identifying it with an object astronomically described as a black hole also requires a separate description of its gravitational and causal properties.

Astrophysical evolutionary models distinguish among the ejection of large amounts of matter, continued infall, and the fallback of material initially moving outward. A black hole can also form when the supernova explosion is weak or fails; subsequent uptake can likewise alter the compact remnant [6]. For the field-mechanical interpretation, this means that changes in embedding must not be described exclusively in terms of ejection. Some matter could be released outward while other matter continues inward, is transformed, or becomes part of an altered central organization. These pathways would have to be brought together within the same energy and motion balance.

The proposed envelope tipping point would therefore be one possible route to exposure, but not a prerequisite for every evolutionary outcome that appears as a black hole. If the effect transferred outward is insufficient for permanent detachment, a temporarily expanded envelope could move inward again. Its fallback would once more alter the supply and coupling conditions of the central region. Conversely, sufficiently effective release could largely remove the previous embedding environment. In both cases, the evolution of the saturated Anun, the evolution of its coupling region, and the fate of the surrounding matter must be distinguished. A bright outburst, extensive exposure, and the formation of a compact central state are therefore not equivalent events.

10.5. Thermonuclear Supernovae as a Distinct Evolutionary Pathway

Type Ia thermonuclear supernovae must be distinguished from core-collapse supernovae. They are associated with the explosive transformation of a white dwarf in a binary system, with mass transfer and the interaction of two white dwarfs among the evolutionary pathways under investigation. Such an explosion can largely or completely destroy the white dwarf involved rather than leave a neutron star or black hole as a compact remnant [6, 7]. This additional pathway shows why the term supernova cannot generally be equated with the exposure of an Anun that survives.

Within UQSH, the starting point here would be a previously compact, maintained matter configuration whose supply or internal coupling is altered through interaction with a companion. The possible transition would involve rapid transformation and dispersal of this host organization. Whether a previously assumed central Anun organization would persist, change, or lose its own maintained form remains unspecified. For this evolutionary pathway in particular, its continued existence must not be inferred solely from the assumption of prior embedding. The field-mechanical description would have to establish which bound contributions persist and which are released in the form of motion, radiation, and dispersed

matter. This does not imply a corresponding supernova pathway for the present-day Sun considered in isolation.

10.6. Shared Conditions and Different Remnant States

In this interpretation, the differences among the pathways described lie in the overall relationship among central organization, material embedding, and the environment. Mass and composition influence the available possibilities for binding and transformation; rotation and directional field organization influence the pathways of motion. Absorbed radiation, spatial differences in tension, and the duration of their action help determine whether changes spread gradually, undergo repeated relief, or lead to rapid rearrangement. It also matters whether a companion supplies additional matter and whether material already moving outward escapes permanently or returns. These conditions are described jointly here without deriving fixed thresholds for the individual evolutionary pathways.

A red giant, a white dwarf, a neutron star, and an exposed Anun therefore do not form a necessary age sequence for the same body. Nor can they be ordered solely by different degrees of a single central saturation. Within the hypothesis, they can represent different forms and evolutionary courses of embedding, supply, and energy exchange. The mathematical connection to *Light in Light Bubbles* provides relations for common dynamics, energy exchange, and the local restoring response [3]. Determining which of these configurations actually emerge from those relations and which persist requires an appropriate combined field state in each case. The present description distinguishes the evolutionary pathways for this purpose and allows for continued embedding as well as partial or extensive exposure.

11. Mathematical Connection to Light in Light Bubbles

11.1. Common Action and Coupled Fields

The field-equation work *Light in Light Bubbles* formulates an effective framework in which orientation, loading, energy exchange, and the gravitational source arise from the same action [3]. This framework is adopted here as an existing mathematical connection. It shows which relations are already described together and where the qualitative e-Anun interpretation connects to them. Applying it to a specific planet, star, or element-formation process subsequently requires mapping the calculated field states to the relevant physical quantities and boundary conditions. The following equations are therefore explained together with their assumptions. Their purpose in this work is to clarify the common organization of the core and its surroundings and the distinction among exchange, saturation, and spatial evolution.

The effective model uses an internal orientation $q \in S^2$ and a loading variable ζ. The orientation describes the local internal alignment of the field organization under consideration; loading is an additional, variable state and exchange channel. Spatial and temporal changes in both quantities contribute to the combined energy and stress. With signature $(+ - - -)$

and $x^0 = ct$, the base action is

$$S_f = \frac{1}{c}\int \sqrt{-g}\,\mathcal{L}_f\,\mathrm{d}^4x, \tag{1}$$

$$\mathcal{L}_f = \frac{AC(\zeta)}{2}\nabla_\mu q\cdot\nabla^\mu q - \frac{\beta_4}{4}J_{\mu\nu}\cdot J^{\mu\nu} + \frac{Z(\zeta)}{2}\nabla_\mu\zeta\nabla^\mu\zeta - V(q) - U(\zeta), \tag{2}$$

$$J_{\mu\nu} = \nabla_\mu q\times\nabla_\nu q, \qquad C(\zeta) = \frac{1}{1+\zeta^2}, \qquad Z(\zeta) = Z_0(1+\zeta^2), \tag{3}$$

$$U(\zeta) = \frac{b_*\zeta^4}{2(1+\zeta^2)}, \qquad V(q) = C_s(1+q_3)^2. \tag{4}$$

Here, $A, \beta_4, Z_0, b_* > 0$ and $C_s \geq 0$. With $P_q = I - qq^\mathsf{T}$ and $X = \nabla_\mu q\cdot\nabla^\mu q$, it follows that

$$P_q\left\{\nabla_\mu\left[AC\nabla^\mu q + \beta_4 J^{\mu\nu}\times\nabla_\nu q\right] + V_q\right\} = 0, \tag{5}$$

$$Z\Box_g\zeta + \frac{Z'}{2}(\nabla\zeta)^2 - \frac{A}{2}C'X + U' = 0. \tag{6}$$

The two equations describe a reciprocal interaction. Through $C(\zeta)$, loading changes the contribution of the orientation organization, while the motion and spatial structure of that organization affect the loading response through X. The projection P_q ensures that the orientation remains on the unit sphere. This reciprocity is essential to the e-Anun interpretation: a changed central state and a changed environment cannot be prescribed independently if both are to arise from the same field organization. The action already contains the relations through which a local change affects other parts of the state. The specific core–envelope configuration that results is determined by the jointly solved profiles and their boundary conditions.

11.2. Mutual Energy Exchange

A central question for the milling function is how an increase in one field contribution is sustained by another contribution or by external supply. In a flat-spacetime comparison, energy contributions ε_q and ε_ζ and the corresponding fluxes $\mathbf{S}_q$ and $\mathbf{S}_\zeta$ can be identified for this purpose. This decomposition assigns the contributions of the two model variables within a common balance. In physical time, their exchange relations are

$$\partial_t\varepsilon_q + \nabla\cdot\mathbf{S}_q = -\mathcal{P}_\zeta, \tag{7}$$

$$\partial_t\varepsilon_\zeta + \nabla\cdot\mathbf{S}_\zeta = \mathcal{P}_\zeta, \tag{8}$$

$$\mathcal{P}_\zeta = \frac{A}{2}C'(\zeta)X\dot{\zeta}. \tag{9}$$

The exchange term cancels from the overall balance:

$$\partial_t\varepsilon + \nabla\cdot\mathbf{S} = 0. \tag{10}$$

The exchange term appears with opposite signs in the two partial balances. It therefore describes an internal transfer that cancels from the overall balance, whereas a flux through the outer boundary can change the energy of the region under consideration. The direction of

transfer depends on the actual field state; the relations allow both transfer and return transfer. This provides a mathematical connection to the assumed recursive transformation. A region can absorb energy and subsequently release it to other contributions or to its surroundings, provided that the coupled dynamics supports that pathway.

The two energy contributions do not yet constitute a complete decomposition into radiation and a particular type of matter. Such an assignment depends on whether the overall organization propagates freely, remains bound, or transitions between these manifestations. The energy balance describes the exchange, while the form and maintenance of the respective state must emerge from the coupled dynamics. For the qualitative e-Anun interpretation, this means that the assumed transformation uses existing energy and that its uptake and release are accounted for together. Internal recursion thus acquires a basis in the balance relations without presupposing a specific elemental yield.

11.3. Bounded Local Restoring Response

For the local static comparison of the base model, without loading derivatives and without an actively adjusted auxiliary channel, the foundational work sets $\chi = \zeta^2$ and $s_q^2 = \sum_i |\partial_i q|^2$. Then

$$e_{\mathrm{loc}} = \frac{A s_q^2 + b_* \chi^2}{2(1+\chi)}, \tag{11}$$

$$\chi_* = \sqrt{1 + A s_q^2 / b_*} - 1, \tag{12}$$

$$e_{\mathrm{adj}} = b_* \left(\sqrt{1 + A s_q^2 / b_*} - 1 \right), \tag{13}$$

$$\mathcal{R}(s_q) = \frac{A s_q}{\sqrt{1 + A s_q^2 / b_*}} \longrightarrow \sqrt{A b_*}. \tag{14}$$

Loading adjusted to the orientation gradient changes the energy cost of further local stress. The expression $\mathcal{R}(s_q)$ approaches a finite limit, while the adjusted energy can continue to increase. What is bounded is therefore the restoring response calculated here with respect to s_q. This distinction supports the e-Anun picture in which a local organizational form can approach saturation while further spatial redistribution or energetic evolution remains possible. It also explains why a local notion of saturation is distinguished from the total energy of a body. By itself, the formula provides neither the power released by an envelope nor an upper bound on all gravitational effects; it describes the specified local comparison within the base model.

A diagnostic value taken from this limiting case is

$$\sigma_{\mathrm{ref}} = \frac{\mathcal{R}}{\sqrt{A b_*}} = \sqrt{\frac{\xi}{1+\xi}}, \qquad \xi = \frac{A s_q^2}{b_*}. \tag{15}$$

For a chosen threshold $0 < \sigma_A < 1$, it follows that

$$\sigma_{\mathrm{ref}} \geq \sigma_A \quad \Longleftrightarrow \quad \xi \geq \frac{\sigma_A^2}{1 - \sigma_A^2}. \tag{16}$$

The diagnostic value compares the local restoring response with its limiting value. A chosen finite threshold thus makes it possible to define a region in which the response is already sufficiently close to this limiting behavior. The choice of threshold is part of the analysis; it is not yet an independently determined universal Anun boundary. Its connection to the regime coordinate Σ_n likewise requires a separate mapping. For the present interpretation, its immediate usefulness is spatial: a specified criterion allows changes in the extent of a highly saturated region to be tracked while the evolution of its total energy is accounted for separately.

11.4. Region Growth and the Maintained State

For a specified saturation indicator,

$$\mathcal{D}_A(t) = \{\mathbf{x} : \sigma(\mathbf{x}, t) \geq \sigma_A\} \tag{17}$$

describes a saturated region to be selected as a connected component. In a radial description, its regular boundary satisfies $\sigma(R_A(t), t) = \sigma_A$, and therefore

$$\dot{R}_A = -\left.\frac{\partial_t \sigma}{\partial_r \sigma}\right|_{r=R_A}, \qquad \partial_r \sigma \neq 0. \tag{18}$$

This relation describes the motion of the specified level surface. Its radius can change if the saturation indicator increases or decreases over time at a fixed location; the spatial gradient determines how that change translates into boundary displacement. This is a kinematic description of the selected boundary. The actual temporal change in σ follows from the coupled field dynamics. For an e-Anun, this makes it possible to distinguish conceptually among an internal state becoming more saturated, expansion of an already highly saturated region, and changes primarily in the surrounding envelope.

In the flat-spacetime comparison, the overall balance for the moving region is

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathcal{D}_A(t)} \varepsilon \,\mathrm{d}V = -\int_{\partial\mathcal{D}_A(t)} \mathbf{S}\cdot\mathbf{n}\,\mathrm{d}A + \int_{\partial\mathcal{D}_A(t)} \varepsilon v_n \,\mathrm{d}A. \tag{19}$$

The second term accounts for the moving boundary. Over a specified time interval, a possible maintained state is characterized by

$$\langle \dot{R}_A \rangle \simeq 0, \qquad \left\langle \frac{\mathrm{d}E_A}{\mathrm{d}t} \right\rangle \simeq 0. \tag{20}$$

The averages refer to a time interval appropriate to the question being considered. Opposing fluxes and internal motion can persist even when the radius and energy of the selected region change little on average. This gives the notion of a maintained planetary e-Anun a more precise meaning: maintenance does not require an absence of motion, but an appropriate joint evolution of the quantities entering the balance. The envelope can continue to supply and absorb energy and matter. Whether the same state is reestablished after a perturbation or a different one develops is a further question of stability, which the average conditions alone do not resolve.

11.5. Regime Participation in the Existing Auxiliary Channel

The latest field-equation work couples a selected scalar regime channel to the same loading potential. Its quasistatic response satisfies

$$\Sigma_n = \tanh[h\delta_{n0} + \kappa(\Sigma_{n-1} + \Sigma_{n+1})]\,, \qquad h = \frac{\eta_\Sigma U(\zeta)}{\mathcal{E}_\Sigma}, \qquad 0 \le \kappa < \frac{1}{2}. \tag{21}$$

For fixed field states, this channel has a unique adjustment. After its elimination,

$$U_{\mathrm{eff}}(\zeta) = U(\zeta) - \mathcal{E}_\Sigma W_\kappa\!\left(\frac{\eta_\Sigma U(\zeta)}{\mathcal{E}_\Sigma}\right) \tag{22}$$

is used in the jointly adjusted model. Here, W_κ denotes the function of the optimized channel response determined in the foundational work; $\mathcal{E}_\Sigma$ and η_Σ are its specified energy scale and coupling parameter. The backreaction is thus accounted for through the same action.

Adjustment of the auxiliary channel changes the common energy description and thereby acts back on the calculated spatial profiles. For the e-Anun interpretation, this provides an existing connection between the participation of supporting regimes and the organization resolved in the model. The selected channel describes a particular contribution under its specified assumptions. Decrease along a regime sequence and spatial decay within the same regime remain distinct processes. Quasistatic adjustment likewise means that the channel response is assumed to establish itself sufficiently rapidly relative to the process under investigation; it is not an independent time-dependent propagation law. Its significance for the qualitative account lies in the common backreaction, through which additional participation can change the organization of the entire calculated state.

11.6. Gravitational Effect of the Overall State

Metric variation of the same action determines the stress–energy tensor. The geometric relation chosen in the foundational work is

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T^{\mathrm{tot}}_{\mu\nu}, \qquad \nabla_\mu T^{\mu\nu}_{\mathrm{tot}} = 0. \tag{23}$$

This geometric relation is retained from the existing field-equation work. The UQSH interpretation concerns the common physical organization from which its source is constructed. This includes the represented contributions from motion, binding, and stress, as well as their interactions. If a nested configuration is resolved more finely, the same energy already included must remain unambiguously assigned in that finer description. The core, envelope, and carrier organization are therefore represented within a consistent overall description. This requirement is particularly important for the e-Anun interpretation because the intense local organization and the extended environment jointly contribute to the external effect.

For a weak, approximately stationary comparison geometry,

$$Q_{\mathrm{tot}} = \varepsilon_{\mathrm{tot}} + \sum_i p_i, \qquad \nabla^2\Psi = \frac{4\pi G}{c^2} Q_{\mathrm{tot}}, \qquad \mathbf{g} = -\nabla\Psi. \tag{24}$$

The principal spatial stresses p_i enter the specified weak-field source combination alongside the energy density. The external effect can therefore depend on changes in organization that are not fully captured by considering energy density alone. For more general states or states that vary strongly over time, the full tensor relation is required. Internal saturation conditions, surrounding matter, and the gravitational effect are thus assessed from the same combined configuration. The mathematical connection supports the qualitative treatment of Anun and environment as an interacting system; its specific identification with a planetary or stellar body remains tied to that body's entire field and matter organization.

12. Spatial Scales and Qualitative Expectations

12.1. Internal Proportions and the Meaning of the Illustrations

For a subsequent quantitative description, the extents of the different regions must be specified separately. The saturation radius R_A denotes the central region selected using a specified criterion. The quantity R_{Fe} describes a possible iron-rich environment, while R_H specifies the extent of the entire host body. An additional range, R_{eff}, is defined using a particular criterion for the effect under consideration. These quantities answer different questions and must therefore not be confused, even in an illustrative representation. The dimensionless ratios

$$\eta_A = \frac{R_A}{R_H}, \qquad \eta_{\mathrm{Fe}} = \frac{R_{\mathrm{Fe}}}{R_H}, \qquad \eta_{\mathrm{eff}} = \frac{R_{\mathrm{eff}}}{R_H} \tag{25}$$

specify which internal proportion is being considered in each case. Here, R_{eff} does not denote a general boundary beyond which gravity ceases. Its meaning depends on which effect is compared against which environmental state.

Planetary and stellar configurations can differ both in their absolute sizes and in these internal ratios. An enlarged detail may therefore be necessary to make a small central region visible within a large envelope. Figure 1 uses the cutaway view to show the e-Anun, its immediate embedding region, and the remaining host matter together. Figure 2, by contrast, illustrates a possible change in embedding and observability. Both images explain the organizational idea. Their radii, colors, and visible interfaces are schematic and do not represent results of a field simulation. For future illustrations drawn to scale, the depicted quantities would first have to be determined from a specified combined state.

12.2. Composition, Maintenance, and Recurring System Families

The qualitative interpretation connects several observable properties that must be considered together. For a planetary body, these include its material composition, internal structure, energy release, and external gravitational effect throughout its evolution. The assumed internal elemental transformation connects the accumulated initial material to the body's subsequent history. An approach that assumes substantial internal transformation must also be able to describe how the associated energy transfer, transport, and maintenance relate to the overall configuration. The notion of a maintained e-Anun provides a framework in which active internal processes and an external appearance that persists on average are considered together.

For stellar configurations, the focus is on the connection among supply, internal motion, envelope organization, and release. When the envelope changes substantially, the preceding central state, the composition of the released material, and subsequent observability must be related to one another. At the level of entire planetary systems, the expectation is that similarities in size ratios and spacing sequences may be associated with comparable formation conditions. The environment is given the same importance as the properties of the central star. This expectation concerns a relationship between conditions and outcomes; a single similarity is not sufficient to attribute its origin to a particular mechanism. The account thus identifies the properties from which further development can proceed without presenting the qualitative explanation as an observational analysis that has already been carried out.

13. Synthesis and Further Connections

In this work, the e-Anun is described as a possible organizational state that connects early embedding, long-term maintenance, and active internal transformation. Its envelope is part of the conditions of its existence and also a region in which the products of reorganization can be transported onward and preserved. This leads to the explicitly planetary extension: within the hypothesis, even a small e-Anun can transform accumulated raw material into heavier elemental organizations within its own central surroundings. The same basic field logic is extended to stellar envelopes and to the broader surroundings of a forming planetary system. Preferred accumulation regions and recurring configurations appear as possible consequences of comparable binding, motion, and exchange conditions, while the evolution of each individual system helps determine its specific form.

The next conceptual step is a jointly described core–envelope state whose spatial regions support different shares of uptake, transformation, and maintenance. The existing field equations provide relations among orientation, loading, energy exchange, and the gravitational source for this purpose. Starting from such states, it becomes possible to investigate progressively how an assumed transformation zone can be connected to a persistent material environment and which additional mappings are required for specific elemental forms. The present version brings together the relevant ideas in detail: spatial differences in tension, absorbed radiation, internal motion, and the duration of their action jointly contribute to evolution. They form the qualitative framework within which the existing mathematics can continue to be used and the new assumptions can be developed in a focused manner.

Author Declarations

Conflicts of interest and funding. The author declares no financial or personal conflicts of interest. This work received no external funding.

Author contribution. Ukshin Q. Rexhepi developed the theoretical framework, formulated the research questions, and is responsible for the interpretation, selection of content, and final approval of the manuscript.

AI and software assistance. AI-assisted and computational tools were used extensively in preparing this manuscript. They supported the organization of the text, the articulation

and linguistic refinement of the ideas developed by the author, the discussion of mathematical relationships, programming and error checking, and the review for repetition and terminological inconsistencies. They were also used to create and revise the LaTeX manuscript.

References

[1] U. Q. Rexhepi. *Phraqtal Field*. UQSH preprint, Version 2, September 2026. doi: 10.5281/zenodo.23290145.

[2] U. Q. Rexhepi. *Gravity-Light*. Universal Quantum Foam Hypothesis. doi: 10.13140/RG.2.2.33455.91042.

[3] U. Q. Rexhepi. *Light in Light Bubbles — Gravitational Light in the Phraqtal Field*. October 26, 2026. doi: 10.5281/zenodo.23652779.

[4] NASA Science. *Types* (Stars). https://science.nasa.gov/universe/stars/types/. Accessed October 28, 2026.

[5] A. I. Karakas and J. C. Lattanzio. *The Dawes Review 2: Nucleosynthesis and Stellar Yields of Low- and Intermediate-Mass Single Stars*. Publications of the Astronomical Society of Australia **31**, e030 (2014). doi: 10.1017/pasa.2014.21.

[6] H.-T. Janka. *Explosion Mechanisms of Core-Collapse Supernovae*. Annual Review of Nuclear and Particle Science **62**, 407–451 (2012). doi: 10.1146/annurev-nucl-102711-094901.

[7] D. Maoz, F. Mannucci, and G. Nelemans. *Observational Clues to the Progenitors of Type Ia Supernovae*. Annual Review of Astronomy and Astrophysics **52**, 107–170 (2014). doi: 10.1146/annurev-astro-082812-141031.